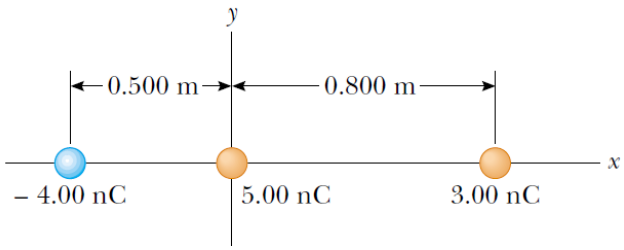
PHYS 1122 EXERCISES

A. Electric Force and Electric Field

1.Three point charges are aligned along the x axis as shown in Figure. Find the electric field at

A. (10 pts) The position (2.00, 0) and,
B. (15 pts) The position (0, 2.00).



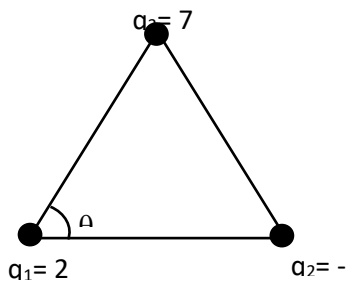
2. An electron and a proton are each placed at rest in an electric field of 520 N/C. Calculate the speed of each particle 48.0 ns after being released.

3. Two identical conducting small spheres are placed with their centers 0.300 m apart. One is given a charge of 12.0 nC and the other a charge of -18.0 nC.

A. Find the electric force exerted by one sphere on the other.
B. If the spheres are connected by a conducting wire, find the electric force each exerts on the other after they have come to equilibrium.

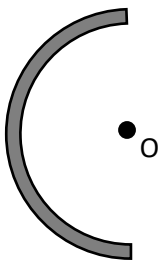
4. Three point charges are placed at the corners of an equilateral triangle of side length 0.500 m and ⍰=60⍰ as shown in the figure. Find:

A. The resultant electric force acting on q3 in unit vector notation
B. The magnitude of the resultant electric force
C. The direction of the resultant electric force



5. A uniformly charged insulating rod of length 14.0 cm is bent into the shape of a semicircle, as shown in Figure. The semicircle rod has a total charge of -7.50 μC.

A. Derive the electric field equation for a such a semi-circle
B. Find the magnitude and direction of the electric field at O, the center of the semicircle.



6. Three solid plastic cylinders all have radius 2.50 cm and length 6.00 cm. Find the charge of each cylinder given the following additional information about each one.

A. *Cylinder 1* carries charge with uniform density 15.0 nC/m^2 everywhere on its surface.

B. *Cylinder 2* carries charge with uniform density 15.0 nC/m^2 on its curved lateral surface only.

C. *Cylinder 3* carries charge with uniform density 500 nC/m3 throughout the plastic.

B. Gauss Law

1. A long, straight metal rod has a radius of 5.00 cm and a charge per unit length of 30.0 nC/m. Find the electric field

A. 3.00 cm,
B. 10.0 cm
C. 100 cm

from the axis of the rod, where distances are measured perpendicular to the rod.

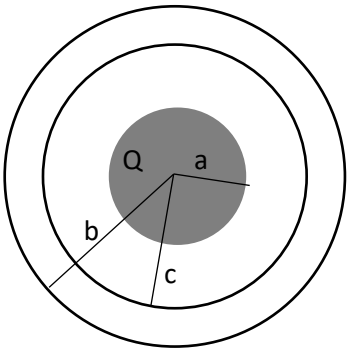
2. A hollow conducting sphere is surrounded by a larger concentric spherical conducting shell. The inner sphere has charge -Q , and the outer shell has net charge +3Q. The charges are in electrostatic equilibrium. Using Gauss's law, find the charges and the electric fields everywhere.

3. Charges of 3.00 nC, -2.00 nC, -7.00 nC, and 1.00 nC are contained inside a rectangular box with length 1.00 m, width 2.00 m, and height 2.50 m. Outside the box are charges of 1.00 nC and 4.00 nC. What is the electric flux through the surface of the box?

4. In the air over a particular region at an altitude of 500 m above the ground, the electric field is 120 N/C directed downward. At 600 m above the ground, the electric field is 100 N/C downward. What is the average volume charge density in the layer of air between these two elevations? Is it positive or negative?

5. Consider a closed triangular box resting within a horizontal electric field of magnitude 7.8x104 N/C, as shown in Figure. Calculate the electric flux through

A. The vertical rectangular surface
B. The slanted surface
C. The entire surface of the box

6. A solid, insulating sphere of radius a has a uniform charge density ρ and a total charge Q. Concentric with this sphere is an uncharged, conducting hollow sphere whose inner and outer radii are b and c, as shown in Figure. Find the magnitude of the electric field in the regions and

A. r < a,
B. a < r < b,
C. r > c